Combinational Gates

X-OR Gate

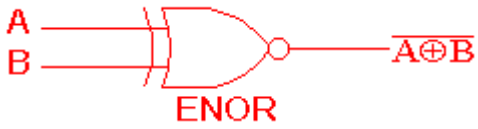
The 'Exclusive-OR' gate is a circuit which will give a high output if either, but not both, of its two inputs are high. An encircled plus sign (⊕) is used to show the X-OR operation.



2 Input EXOR gate		
A	B	A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

X-NOR Gate

The 'Exclusive-NOR' gate circuit does the opposite to the X-OR gate. It will give a low output if either, but not both, of its two inputs are high. The symbol is an X-OR gate with a small circle on the output. The small circle represents inversion.



2 Input EXNOR gate		
A	B	$\overline{A⊕B}$
0	0	1
0	1	0
1	0	0
1	1	1